

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [70, 69, 75, 68, 66, 75, 70, 75, 81, 83]

Tap amplitudes (pixels): [207.19314660480447, 187.44865963777923, 170.53152201279386, 173.3666634621547, 170.4494059831245, 164.83021567661677, 176.58142597679972, 182.87700784953805, 167.70509831248424, 176.7766952966369, 174.6682569902156]

Gemini 3 Pro

Based on the extracted features provided, here is the analysis and scoring for MDS-UPDRS Item 3.4:

Analysis

1.

Rhythm and Speed (Tap Periods):

- The tap periods are: [70, 69, 75, 68, 66, 75, 70, 75, 81, 83]
- There is some variability in the rhythm (e.g., fluctuating between 66 and 75 frames).
- There is a noticeable slowing trend at the very end of the sequence (increasing to 81 and 83 frames).
- This behavior aligns with **Score 1** (Slight slowing/interruptions) or potentially **Score 2** (Mild slowing).

2.

Amplitude (Tap Amplitudes):

- The tap amplitudes are: [207.2, 187.4, 170.5, 173.4, 170.4, 164.8, ...]
- The first tap is the largest (207.2 pixels).
- There is an immediate drop in amplitude for the second tap (187.4) and the third tap (170.5).
- The amplitude continues to trend downwards until the 6th tap (164.8) and never recovers to the initial height of the first tap.

Scoring Application

The scoring criteria state that the score is determined by "Any of the following" conditions being met. We evaluate the amplitude decrement timing:

- **Score 1:** Amplitude decrements near the end.
- **Score 2:** Amplitude decrements midway.
- **Score 3:** Amplitude decrements starting after the 1st tap.

The data shows a clear decrement starting immediately after the 1st tap (dropping from 207 to 187 to 170). Because this specific criterion for **Score 3** is met, it dictates the final score, regardless of the milder slowing observed in the rhythm.